



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA	Div: B	Course Code: MCA01604	Batch: S2
Semester: III		Course Name: Data Science Laboratory	
Name: Tejashree Vijay Dumasia		Roll No: 52117	
CO No: CO605.1		Assignment No: 04	

Title: A company wants to predict the salary of employees based on their years of experience. The company has collected data on the salaries and experience of 100 employees. Use simple linear regression to model the relationship between salary and experience

Code:

```
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt

# Generate synthetic data for 100 employees
np.random.seed(42)
years_experience = np.random.uniform(1, 20, 100)
# Assume salary = 3000 * years_experience + noise
salaries = 3000 * years_experience + np.random.normal(0, 5000, 100)

df = pd.DataFrame({'YearsExperience': years_experience, 'Salary': salaries})

# Split data into train and test sets
X = df[['YearsExperience']]
y = df['Salary']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train linear regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict on test set
y_pred = model.predict(X_test)

# Plot results
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X_test, y_pred, color='red', linewidth=2, label='Regression Line (Test)')
plt.xlabel('Years of Experience')
```

```
plt.ylabel('Salary')  
plt.title('Salary vs. Years of Experience (Linear Regression)')  
plt.legend()  
plt.show()
```

```
# Print model coefficients  
print(f"Intercept: {model.intercept_:.2f}")  
print(f"Coefficient: {model.coef_[0]:.2f}")
```

Output:

```
Intercept: 820.19  
Coefficient: 2894.38
```

